

Chapter 4, Section 4

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3. Let the elliptic curve X be embedded in $\mathbb{P}^2$ so as to have the equation $y^2 = x(x-1)(x-\lambda)$. Show that any automorphism of X leaving $P_0 = (0 : 1 : 0)$ fixed is induced by an automorphism of $\mathbb{P}^2$ coming from the automorphism of the affine (x, y) -plane given by

$$\begin{cases} x' = ax + b \\ y' = cy. \end{cases}$$

In each of the four cases of (4.7), describe these automorphisms of $\mathbb{P}^2$ explicitly, and hence determine the structure of the group $G = \text{Aut}(X, P_0)$.

Proof. Let $f : X \rightarrow \mathbb{P}^1$ be the unique g_2^1 with $f(P_0) = \infty$, branched over $0, 1, \lambda, \infty$. If $\sigma \in G$, then by (4.4) there exists an automorphism τ of $\mathbb{P}^1$ which fixes ∞ and satisfies $f \circ \sigma = \tau \circ f$. In particular, τ sends $\{0, 1, \lambda\}$ to $\{0, 1, \lambda\}$ in some order. If $\tau = \text{id}$, then either $\sigma = \text{id}$ or σ is the unique automorphism of order 2 fixing P_0 and interchanging the sheets of f . We call it the *hyperbolic involution* of X and denote it by ι . It comes from the automorphism of the affine (x, y) -plane corresponding to the reflection across the x -axis:

$$\begin{cases} x' = x \\ y' = -y. \end{cases}$$

The other automorphisms are determined using the methods of (4.7). In particular, we look at the orbits of the action of S_3 on $k - \{0, 1\}$ (4.5).

$j \neq 0, 1728$ Since the orbits of λ are all distinct, $\tau = \text{id}$, and $\sigma = \text{id}$ or ι .

$j = 1728, \text{char } k \neq 3$ This is the case $\lambda = -1, \frac{1}{2}$, or 2 , and λ coincides with one other element of its orbit under the action of S_3 . For example, if $\lambda = -1$, then $\lambda = 1/\lambda$, and G is the cyclic group of order 4 generated by

$$\begin{cases} x' = -x \\ y' = iy. \end{cases}$$

$j = 0, \text{char } k \neq 3$ This is the case $\lambda = -\omega$ or $-\omega^2$, where ω is a third root of unity, and λ coincides with two other elements of its orbit under S_3 . For example, if $\lambda = -\omega$, then

$$\lambda = \frac{1}{1-\lambda} = \frac{\lambda-1}{\lambda},$$

and G is the cyclic group of order 6 generated by

$$\begin{cases} x' = \omega^2 x + 1 \\ y' = -y. \end{cases}$$

$j = 0 (= 1728), \text{char } k = 3$ If $\text{char } k = 3$ and $j = 0$, then $\lambda = -1$, and all six elements of its orbit under S_3 coincide. Note that k has no nontrivial third roots of unity, and the values $\lambda = -1, \frac{1}{2}, 2$ of the previous case coincide too. So G is the cyclic group of order 12 generated by

$$\begin{cases} x' = x + 1 \\ y' = iy. \end{cases}$$

□

4. Let X be an elliptic curve in $\mathbb{P}^2$ given by an equation of the form

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6.$$

Show that the j -invariant is a rational function of the a_i , with coefficients in $\mathbb{Q}$. In particular, if the a_i are all in some field $k_0 \subseteq k$, then $j \in k_0$ also. Furthermore, for every $\alpha \in k_0$, there exists an elliptic curve defined over k_0 with j -invariant equal to α .

Proof. Complete the square in y to get an equation of the form

$$y^2 = x^3 + Ax^2 + Bx + C,$$

with

$$A = \frac{a_1^2}{4} + a_2, \quad B = \frac{a_1a_3}{2} + a_4, \quad C = \frac{a_3^2}{4} + a_6.$$

Let α, β, γ be the distinct roots of the cubic on the right-hand side. Applying a linear transformation of x , which sends α, β to $0, 1$, transforms the equation of X into

$$y^2 = x(x-1)(x-\lambda), \quad \lambda = \frac{\gamma-\alpha}{\beta-\alpha}.$$

We can plug in λ into the j -invariant formula. Simplifying, we get a rational function of the elementary symmetric polynomials of α, β, γ (which are just A, B, C) with coefficients in $\mathbb{Q}$

$$j = 2^8 \frac{(A^2 - 3B)^3}{A^2B^2 - 4B^3 - 4A^3 + 18ABC - 27C^2}.$$

Since A, B, C are polynomials of the a_i with integer coefficients, the j -invariant is a rational function of the a_i with coefficients in $\mathbb{Q}$. $\square$

5. Let X, P_0 be an elliptic curve having an endomorphism $f : X \rightarrow X$ of degree 2.

- If we represent X as a 2-1 covering of $\mathbb{P}^1$ by a morphism $\pi : X \rightarrow \mathbb{P}^1$ ramified at P_0 , then as in (4.4), show that there is another morphism $\pi' : X \rightarrow \mathbb{P}^1$ and a morphism $g : \mathbb{P}^1 \rightarrow \mathbb{P}^1$, also of degree 2, such that $\pi \circ f = g \circ \pi'$.
- For a suitable choice of coordinates in the two copies of $\mathbb{P}^1$, show that g can be taken to be the morphism $x \mapsto x^2$.
- Now show that g is branched over two of the branch points of π , and that g^{-1} of the other two branch points of π consists of the four branch points of π' . Deduce a relation involving the invariant λ of X .
- Solving the above, show that there are just three values of j corresponding to elliptic curves with an endomorphism of degree 2, and find the corresponding values of λ and j .

Proof.

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$\square$

- Let X be a curve of genus g embedded birationally in $\mathbb{P}^2$ as a curve of degree d with r nodes. Generalize the method of (Ex. 2.3) to show that X has $6(g-1) + 3d$ inflection points. A node does not count as an inflection point. Assume $\text{char } k = 0$.
 - Now let X be a curve of genus g embedded as a curve of degree d in $\mathbb{P}^n$, $n \geq 3$, not contained in any $\mathbb{P}^{n-1}$. For each point $P \in X$, there is a hyperplane H containing P , such that P counts at least n times in the intersection $H \cap X$. This is called an *osculating hyperplane* at P . It generalizes the notion of tangent line for curves in $\mathbb{P}^2$. If P counts at least $n+1$ times in $H \cap X$, we say H is a *hyperosculating hyperplane*, and that P is a *hyperosculating point*. Use Hurwitz's theorem as above, and induction on n , to show that X has $n(n+1)(g-1) + (n-1)d$ hyperosculating points.

- (c) If X is an elliptic curve, for any $d \geq 3$, embed X as a curve of degree d in $\mathbb{P}^{d-1}$, and conclude that X has exactly d^2 points of order d in its group law.

Proof.

- (a) Let S be the singular locus of $X \subset \mathbb{P}^2$, which consists of r nodes. The dual curve $X^* \subset (\mathbb{P}^2)^*$ can be defined for $X - S$. In particular, $X - S$ has $3d(d-2)$ inflection points, which can be rewritten as
- (b)
- (c)

□

7. The Dual of a Morphism. Let X and X' be elliptic curves over k , with base points P_0, P'_0 .

- (a) If $f : X \rightarrow X'$ is any morphism, use (4.11) to show that $f^* : \text{Pic } X' \rightarrow \text{Pic } X$ induces a homomorphism $\hat{f} : (X', P'_0) \rightarrow (X, P_0)$. We call this the *dual* of f .
- (b) If $f : X \rightarrow X'$ and $g : X' \rightarrow X''$ are two morphisms, then $\widehat{g \circ f} = \hat{f} \circ \hat{g}$.
- (c) Assume $f(P_0) = P'_0$, and let $n = \deg f$. Show that if $Q \in X$ is any point, and $f(Q) = Q'$, then $\hat{f}(Q') = n_X(Q)$. Conclude that $f \circ \hat{f} = n_{X'}$ and $\hat{f} \circ f = n_X$.
- (d) If $f, g : X \rightarrow X'$ are two morphisms preserving the base points P_0, P'_0 , then $\widehat{f+g} = \hat{f} + \hat{g}$.
- (e) Using (d), show that for any $n \in \mathbb{Z}$, $\hat{n}_X = n_X$. Conclude that $\deg n_X = n^2$.
- (f) Show for any f that $\deg \hat{f} = \deg f$.

Proof. Let the pair $(X, \mathcal{P})$ (resp. $(X', \mathcal{P}')$) be the Jacobian of (X, P_0) (resp. (X', P'_0)). If $\mathcal{C}$ is any category, and T is any object in $\mathcal{C}$, then denote by $h_T : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ the contravariant Hom functor defined by $U \mapsto \text{Hom}_{\mathcal{C}}(U, T)$. Recall that X and $\mathcal{P} \in \text{Pic}^\circ(X/X)$ satisfies the following universal property: if T is any scheme of finite type over k , and for any $\mathcal{M} \in \text{Pic}^\circ(X/T)$, there is a unique morphism $f : T \rightarrow X$ such that $f^*\mathcal{P} \cong \mathcal{M}$ in $\text{Pic}^\circ(X/T)$. In other words, $(X, \mathcal{P})$ represents the functor $\text{Pic}^\circ(X/-)$. Lastly, since $(X, \mathcal{P})$ is a group object in the category of schemes of finite type over k , the set $h_X(T)$ has a group structure induced by $\text{Pic}^\circ(X/T)$.

- (a) Any morphism $f : X \rightarrow X'$ defines a natural transformation $f^* : \text{Pic}^\circ(X'/-) \rightarrow \text{Pic}^\circ(X/-)$ of group functors in the following way (4.10.2).

If T is any scheme of finite type over k , then $f' = f \times 1_T : X' \times T \rightarrow X \times T$ induces a group homomorphism $f^* : \text{Pic}(X' \times T) \rightarrow \text{Pic}(X \times T)$. Let $p : X \times T \rightarrow T$ and $p' : X' \times T \rightarrow T$ be the projection maps. The image of f^* lies in $\text{Pic}^\circ(X \times T)$, and $p = p' \circ f'$ implies $f^*p'^*\text{Pic}(T) = p^*\text{Pic}(T)$. Hence, we have a well-defined group homomorphism $f_T^* : \text{Pic}^\circ(X'/T) \rightarrow \text{Pic}^\circ(X/T)$ for each scheme T of finite type over k .

It remains to show f^* is functorial. Let $g : S \rightarrow T$ be any morphism of schemes of finite type over k . We can apply $\text{Pic}(-)$ to the following commutative diagram

$$\begin{array}{ccc} X \times S & \xrightarrow{f \times 1_S} & X' \times S \\ 1_X \times g \downarrow & & \downarrow 1_{X'} \times g \\ X \times T & \xrightarrow{f \times 1_T} & X' \times T \end{array}$$

which gives a commutative diagram of Picard groups with arrows given by taking inverse images of invertible sheaves.

Now, we identify h_X and $h_{X'}$ with $\text{Pic}^\circ(X/-)$ and $\text{Pic}^\circ(X'/-)$, respectively. Yoneda's lemma says there exists a natural bijection between the set of morphisms $X' \rightarrow X$ and the set of natural transformations of functors $h_{X'} \rightarrow h_X$. Hence, there exists a unique morphism $\hat{f} \in \text{Hom}(X', X)$ such that $f^* \in \text{Hom}(h_{X'}, h_X)$ is given by composition with $\hat{f}$. The invertible sheaves $\mathcal{O}_X$ and $\mathcal{O}_{X'}$ correspond to $P_0 \in X$ and $P'_0 \in X'$, respectively, and structure sheaves pullback, so we conclude that $\hat{f}(P'_0) = P_0$.

- (b) This is a consequence of the uniqueness property of $\hat{f}$ in $\text{Hom}(X', X)$, and the analogous fact for inverse image of sheaves, i.e., $(g \circ f)^* = f^* \circ g^*$.

- (c) The points $Q' \in X'$ and $\hat{f}(Q') \in X$ correspond to $\mathcal{O}_{X'}(Q' - P'_0)$ in $\text{Pic}^\circ(X'/k)$ and $f^*\mathcal{O}_{X'}(Q' - P'_0)$ in $\text{Pic}^\circ(X/k)$, respectively. Thus, it is enough to show the linear equivalence of the following divisors

$$f^*(Q' - P'_0) \sim n_X(Q) - P_0.$$

By the hypothesis, $Q' - P'_0 = f_*(Q - P_0)$, which by (Ex. 2.6) implies

$$f^*(Q' - P'_0) = n(Q - P_0).$$

The right-hand side is linearly equivalent to $n_X(Q) - P_0$ because the group law on X is induced by that of $\text{Pic}^\circ(X/k)$, so we conclude that $\hat{f} \circ f = n_X$. For the second corollary, observe that f and multiplication by n commutes since f is a morphism of proper group schemes. Hence, we have

$$(f \circ \hat{f})(Q') = f(n_X(Q)) = n_{X'}(f(Q)) = n_{X'}(Q').$$

- (d) It is enough to show for any $\mathcal{L}' \in \text{Pic}^\circ(X')$, that

$$(f + g)^*\mathcal{L}' \cong f^*\mathcal{L}' \otimes g^*\mathcal{L}'.$$

Let Q' be the closed point in X' such that $\mathcal{L}' \cong \mathcal{O}_{X'}(Q' - P'_0)$. There exists a bijection

$$\begin{aligned} \text{Hom}_{\text{Sch}(k)}(X, X') &\xrightarrow{\sim} \text{Pic}^\circ(X'/X) \\ f : X &\rightarrow X' \mapsto \mathcal{L}_f := (f \times 1_{X'})^*\mathcal{P}', \end{aligned}$$

where the group law on the left-hand side is induced by $\text{Pic}^\circ(X'/X)$, i.e., $\mathcal{L}_{f+g} = \mathcal{L}_f \otimes \mathcal{L}_g$ in $\text{Pic}(X \times X')$ for any $f, g \in h_{X'}(X)$. Let $\sigma : X \rightarrow X \times X'$ be the section $x \mapsto (x, Q')$. If we assume $\mathcal{P}' = \mathcal{O}_{X' \times X'}(\Delta - X' \times P'_0)$ as in (4.11), then $\sigma^*\mathcal{L}_f \cong f^*\mathcal{O}_{X'}(Q')$, so replacing $\mathcal{P}'$ by $\mathcal{P}' \otimes p_2^*\mathcal{O}_{X'}(-P_0)$, we get $\sigma^*\mathcal{L}_f \cong f^*\mathcal{L}'$. Hence,

$$(f + g)^*\mathcal{L}' \cong \sigma^*\mathcal{L}_{f+g} \cong \sigma^*\mathcal{L}_f \otimes \sigma^*\mathcal{L}_g \cong f^*\mathcal{L}' \otimes g^*\mathcal{L}'.$$

- (e) Proceeding by induction on n reduces the statement to the case $n = 2$. The dual of the identity is the identity. Hence, $\hat{2}_X = \widehat{1_X + 1_X} = 1_X + 1_X = 2_X$ by (d). According to (c), $\hat{n}_X \circ n_X$ is multiplication by $\deg n_X$. Since $\hat{n}_X \circ n_X = n_X^2$, we conclude that $\deg n_X = n^2$.
- (f) By (c) and (e), $\deg \hat{f} \circ f = n^2$. The degree of morphisms of curves is multiplicative with respect to composition. Hence, $\deg \hat{f} = n$.

□

9. We say two elliptic curves X, X' are *isogenous* if there is a finite morphism $f : X \rightarrow X'$.

- (a) Show that isogeny is an equivalence relation.
- (b) For any elliptic curve X , show that the set of elliptic curves X' isogenous to X , up to isomorphism is countable.

Proof.

- (a) The identity morphism is trivially finite, so any elliptic curve is isogenous to itself. A composition of finite morphisms is finite, so isogeny satisfies transitivity. Lastly, (Ex. 4.7) implies isogeny is reflexive. Hence, isogeny is an equivalence relation.
- (b) Since $\ker f$ is a closed subgroup scheme of X , it is a finite set of points, all of which are torsion under the group law of X . Every torsion point of X lies in the kernel of a multiplication map $n_X : X \rightarrow X$ for some $n > 0$. By (Ex. 4.7), n_X is an étale covering of degree n^2 , which implies X has exactly n^2 points of order n . Hence, there are countably many torsion points on X . The set of finite subsets of a countable set is countable, so there are only countably many candidates for $\ker f$. Since X' is uniquely determined by X and $\ker f$, the set of elliptic curves isogenous to X is countable.

□

11. Let X be an elliptic curve over $\mathbb{C}$, defined by the elliptic functions with periods $1, \tau$. Let R be the ring of endomorphisms of X .

- (a) If $f \in R$ is a nonzero endomorphism corresponding to complex multiplication by α as in (4.18), show that $\deg f = |\alpha|^2$.
- (b) If $f \in R$ corresponds to $\alpha \in \mathbb{C}$ again, show that the dual $\hat{f}$ of (Ex. 4.7) corresponds to the complex conjugate $\bar{\alpha}$ of α .
- (c) If $\tau \in \mathbb{Q}(\sqrt{-d})$ happens to be integral over $\mathbb{Z}$, show that $R = \mathbb{Z}[\tau]$.

Proof.

- (a)
- (b) By (Ex. 7c), $\hat{f} \circ f$ is multiplication by $|\alpha|^2$, but f is multiplication by α , and $\bar{\alpha}$ is the unique element in $\mathbb{C}$ such that $\alpha\bar{\alpha} = |\alpha|^2$. Hence, $\hat{f}$ is multiplication by $\bar{\alpha}$.
- (c)

□

12. Again let X be an elliptic curve over $\mathbb{C}$ determined by the elliptic functions with periods $1, \tau$, and assume that τ lies in the region G of (4.15B).

- (a) If X has any automorphisms leaving P_0 fixed other than ± 1 , show that either $\tau = i$ or $\tau = \omega$, as in (4.20.1) and (4.20.2). This gives another proof of the fact (4.7) that there are only two curves, up to isomorphism, having automorphisms other than ± 1 .
- (b) Now show that there are exactly three values of τ for which X admits an endomorphism of degree 2. Can you match these with the three values of j determined in (Ex. 4.5)?

Proof.

- (a)
- (b)

□

13. If $p = 13$, there is just one value of j for which the Hasse invariant of the corresponding curve is 0. Find it.

Proof. The polynomial $h_p(\lambda)$ of (4.22) for $p = 13$ is

$$h_{13}(\lambda) = \lambda^6 + 10\lambda^5 + 4\lambda^4 + 10\lambda^3 + 4\lambda^2 + 10\lambda + 1,$$

and an elliptic curve has Hasse invariant 0 if and only if $h_{13}(\lambda) = 0$. On the otherhand, consider the j -invariant

$$j = 2^8 \frac{(\lambda^2 - \lambda + 1)^3}{\lambda^2(\lambda - 1)^2}.$$

Expanding above modulo 13 gives

$$\lambda^6 + 10\lambda^5 + (10j + 6)\lambda^4 + (6j + 6)\lambda^3 + (10j + 6)\lambda^2 + 10\lambda + 1 = 0.$$

Matching coefficients with $h_{13}(\lambda)$ and solving for j gives $j \equiv 5 \pmod{13}$.

□

14. The Fermat curve $X: x^3 + y^3 = z^3$ gives a nonsingular curve in characteristic p for every $p \neq 3$. Determine the set $\mathfrak{P} = \{p \neq 3 \mid X_{(p)} \text{ has Hasse invariant } 0\}$, and observe (modulo Dirichlet's theorem) that it is a set of primes of density $1/2$.

Proof. Using (4.21) with $f(x, y, z) = x^3 + y^3 - z^3$, we find that the coefficient of $(xyz)^{p-1}$ in f^{p-1} is zero if and only if $p \equiv 2 \pmod{3}$. By Dirichlet's theorem, the density of $\mathfrak{P}$ is $1/\phi(3) = 1/2$. tunnel is formed in

□

- 15.** Let X be an elliptic curve over a field k of characteristic p . Let $F' : X_p \rightarrow X$ be the k -linear Frobenius morphism (2.4.1). Use (4.10.7) to show that the dual morphism $\hat{F}' : X \rightarrow X_p$ is separable if and only if the Hasse invariant of X is 1. Now use (Ex. 4.7) to show that if the Hasse invariant is 1, then the subgroup of points of order p on X is isomorphic to $\mathbb{Z}/p$; if the Hasse invariant is 0, then it is 0.

Proof.

□

- 16.** Again let X be an elliptic curve over k of characteristic p , and suppose X is defined over the field $\mathbb{F}_q$ of $q = p^r$ elements, i.e., $X \subseteq \mathbb{P}^2$ can be defined by an equation with coefficients in $\mathbb{F}_q$. Assume also that X has a rational point over $\mathbb{F}_q$. Let $F' : X_q \rightarrow X$ be the k -linear Frobenius with respect to q .

- Show that $X_q \cong X$ as schemes over k , and that under this identification, $F' : X_q \rightarrow X$ is the map obtained by the q th-power map on the coordinates of points of X , embedded in $\mathbb{P}^2$.
- Show that $1_X - F'$ is a separable morphism and its kernel is just the set $X(\mathbb{F}_q)$ of points of X with coordinates in $\mathbb{F}_q$.
- Using (Ex. 4.7), show that $F' + \hat{F}' = a_X$ for some integer a , and that $N = q - a + 1$, where $N = \#X(\mathbb{F}_q)$.
- Use the fact that $\deg(m + nF') > 0$ for all $m, n \in \mathbb{Z}$ to show that $|\alpha| \leq 2\sqrt{q}$. This is Hasse's proof of the analogue of the Riemann hypothesis for elliptic curves.
- Now assume $q = p$, and show that the Hasse invariant of X is 0 if and only if $a \equiv 0 \pmod{p}$. Conclude that for $p \geq 5$ that X has Hasse invariant 0 if and only if $N = p + 1$.

Proof.

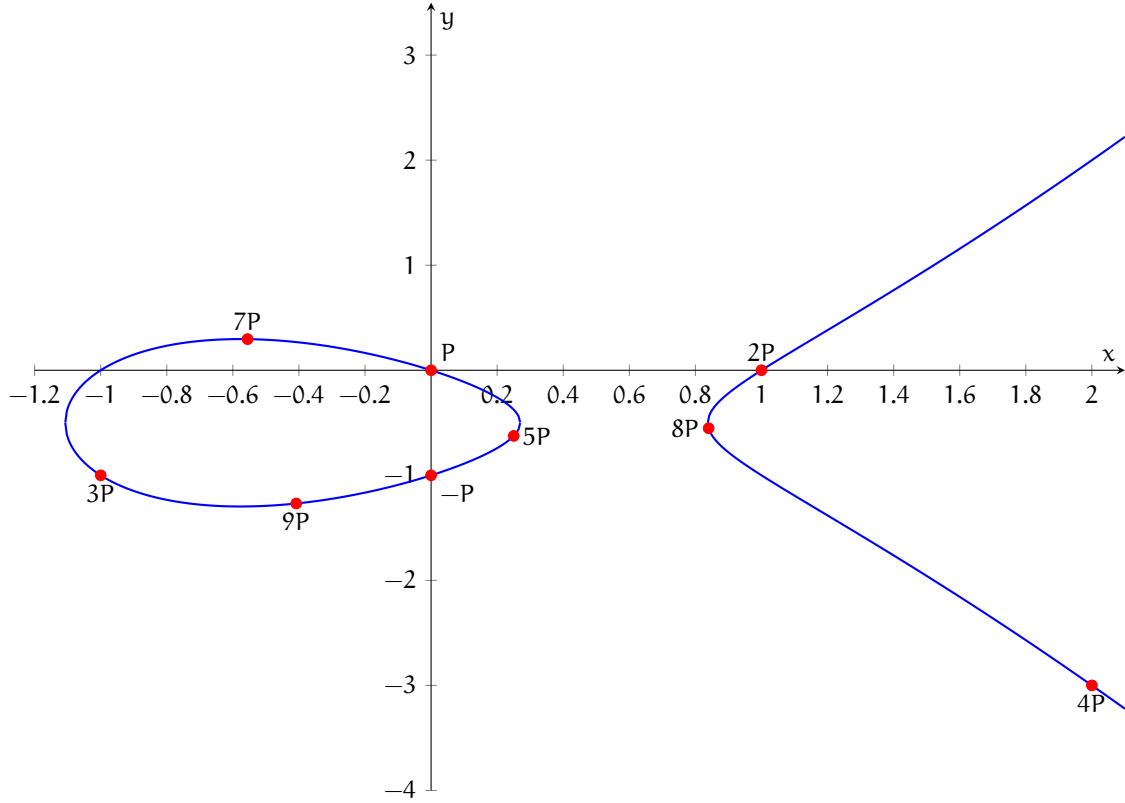
□

- 17.** Let X be the curve $y^2 + y = x^3 - x$ of (4.23.8).

- If $Q = (a, b)$ is a point on the curve, compute the coordinates of the point $P + Q$, where $P = (0, 0)$, as a function of a, b . Use this formula to find the coordinates nP , $n = 1, 2, \dots, 10$.
- This equation defines a nonsingular curve over $\mathbb{F}_q$ for all $p \neq 37$.

Proof.

$$(a) \ P + Q = \left(-\frac{a+b}{a^2}, \frac{b(a+b)}{a^3} - 1 \right), \quad -P = (0, -1), \quad 2P = (1, 0)$$



(b) Let X be the cubic curve in $\mathbb{P}^2$ given by an equation of the form

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6.$$

Define the following quantities

$$b_2 = a_1^2 + 4a_2$$

$$b_4 = 2a_4 + a_1a_3$$

$$b_6 = a_3^2 + 4a_6$$

$$b_8 = a_1^2a_6 + 4a_2a_6 - a_1a_3a_4 + a_2a_3^2 - a_4^2.$$

The discriminant of X is given by

$$\Delta = -b_2^2b_8 - 8b_4^3 - 27b_6^2 + 9b_2b_4b_6,$$

and X defines a nonsingular curve over $\mathbb{F}_q$ if and only if $\Delta \neq 0$ in $\mathbb{F}_q$. For our elliptic curve, the quantities b_i are

$$b_2 = 0, \quad b_4 = -2, \quad b_6 = 1, \quad b_8 = -1,$$

so the determinant of X is $\Delta = 37$. Hence, X defines a nonsingular curve over $\mathbb{F}_q$ for all $p \neq 37$. □

20. Let X be an elliptic curve over a field k of characteristic $p > 0$, and let $R = \text{End}(X, P_0)$ be its ring of endomorphisms.

(a) Let X_p be the curve over k defined by changing the k -structure of X (2.4.1). Show that $j(X_p) = j(X)^{1/p}$. Thus, $X \cong X_p$ over k if and only if $j \in \mathbb{F}_p$.

(b) Show that p_X in R factors into a product of $\pi\hat{\pi}$ of two elements of degree p if and only if $X \cong X_p$. In this case, the Hasse invariant of X is 0 if and only if π and $\hat{\pi}$ are associates in R .

(c) If $\text{Hasse}(X) = 0$ show in any case $j \in \mathbb{F}_{p^2}$.

- (d) For any $f \in R$, there is an induced map $f^* : H^1(\mathcal{O}_X) \rightarrow H^1(\mathcal{O}_X)$. This must be multiplication by an element $\lambda_f \in k$. So we obtain a ring homomorphism $\varphi : R \rightarrow k$ by sending f to λ_f . Show that any $f \in R$ commutes with the (nonlinear) Frobenius morphism $F : X \rightarrow X$, and conclude that if $\text{Hasse}(X) \neq 0$, then the image of φ is $\mathbb{F}_p$. Therefore, R contains a prime ideal $\mathfrak{p}$ with $R/\mathfrak{p} \cong \mathbb{F}_p$.

Proof.

- (a)
- (b)
- (c)
- (d)

□

22. If $X \rightarrow \mathbb{A}_{\mathbb{C}}^1$ is a family of elliptic curves having a section, show that the family is trivial.

Proof.

□